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MILITARY SPECIFICATION

INSIGNIA, EMBROIDERED, ORGANIZATIONAL INSIGNIA AND BADGES, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification establishes the requirements for Schiffli embroidered organizational insignia, badges and branch of service insignia.

1.2 Classification. The insignia covered by this specification shall be of the following types, styles and classes as specified (See 6.2).

Type I - Organizational Insignia

Style 1 - Army
Style 2 - Air Force
Class 1 - Full Color
Class 2 - Subdued

Type II - Badges

Style 1 - Army
Style 2 - Air Force
Class 1 - Ultramarine Blue Base Cloth
Class 2 - Olive Green Base Cloth

Type III - Insignia, Branch of Service

Style 1 - Army
Style 2 - Air Force
Class 1 - Ultramarine Blue Base Cloth
Class 2 - Olive Green Base Cloth

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, U.S. Army, Cameron Station, Alexandria, VA 22304-5050 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
UU-P 268 - Paper, Kraft, Wrapping
CCC-C-461 - Cloth, Twill, Uniform Cotton
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-P-15011 - Pallet, Material Handling, Wood, Post Construction, 4 Way entry

STANDARDS

FEDERAL

FED-STD-191 - Textile, Test Methods
FED-STD-751 - Stitches, Seams and Stitching

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 Thread, Cotton

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A-A-50199 Thread, Polyester Core, Cotton or Polyester Covered

DRAWINGS

THE INSTITUTE OF HERALDRY

A-1	(Series)	Army Shoulder Sleeve Insignia, Various
A-4	(Series)	Beret Flash, Various
A-6	(Series)	Background Trimmings, Various
A-10J	(Series)	Shoulder Sleeve Insignia, JROTC, Various
A-10S	(Series)	Shoulder Sleeve Insignia, SROTC, Various
A-13	(Series)	Badges, Identification, Various
B-2	(Series)	Insignia, Branch of Service, Various
B-5	(Series)	Badges, Identification, Various
B-15	(Series)	USAF Command/Organizational Patches, Various

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

THE COLOR ASSOCIATION OF THE UNITED STATES

The Standard Color Reference of America, Tenth Edition.

United States Army Color Card, The Official Colors for Branches, Agencies, etc.

DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1974.

(Application for copies of the color cards should be addressed to the Color Association of the United States., 343 Lexington Avenue, New York, NY 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Standard Practice for Commercial Packaging
ASTM-D-5034 - Breaking Force and Elongation of Textile Fabrics (Grab Test)

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Unless otherwise specified (see 6.3), samples shall be furnished by the contracting officer to be used for guidance, information and as a standard for colorfastness. In the event there is a variation between this specification and the guide sample, the specification shall govern.

3.2 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material. Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Base cloth. The base cloth shall be 100 percent cotton, 50/50 polyester/cotton or 65/35 polyester/cotton blend, three up, one down, left hand twill, 7.0 to 8.3 ounces per square yard, single ply warp and filling, with a breaking strength of not less than 150 lbs in the warp and 80 lbs in the filling. The residual shrinkage shall be not more than 2 percent in either the warp or filling direction and the cloth shall have not less than fair colorfastness to laundering when tested as specified in 4.5.2.1.

3.3.1.1 Color. The color of the base cloth shall be as follows unless otherwise specified on the applicable drawing or cartoon.

Type	Color of embroidered design edge	Color of base cloth
Type I, Style 1 Class 1	Shades of maroon and red	Khaki
	White	White
	Ultramarine blue	Ultramarine blue
	All other blues and purple	Navy blue
	All shades of black, brown and green	Army green 159
	All shades of yellow, orange, tan and gray or combination of colors	Khaki
Type I, Style 2 Class 1	All insignia	Khaki
Type II and III, Style 2, Class 1	All insignia	Ultramarine blue
Type I, II and III, Style 1 and 2, Class 2	All insignia	Olive green 107

3.3.2 Buckram. The buckram or backing material, when specified, shall be a light, open-weave, single ply fabric weighing 2.5 ounces (minimum) per square yard having a minimum thread count of 40 ends in the warp and 20 ends in the filling. The

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buckram shall have a minimum breaking strength of 30 pounds in the warp direction and 20 pounds in the filling direction. Testing shall be in accordance with 4.5.2.1.

3.3.3 Embroidery yarns. The embroidery yarn shall be a semi-dull continuous filament, regenerated cellulose type rayon, two-ply schiffli type yarn. The yarn shall be 150, 100 or 75 denier as specified on the applicable cartoon or drawing. Unless otherwise specified, the use of polyester is authorized in lieu of rayon. When polyester yarn is used, 135 denier shall be substituted for 150 denier, 90 denier shall be substituted for 100 denier and 70 denier shall be substituted for 75 denier rayon.

3.3.3.1 Color. The color shall be as specified on the applicable drawing or cartoon. When overedge stitching is required, the color shall be the same as the embroidered border. The use of sulphur dyes is prohibited.

3.3.3.2 Matching. The color shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 plus or minus 500 K, with illumination of 100 plus or minus 20 foot candles and shall be a good match to the sample under incandescent lamplight at 2300 plus or minus 100K. The degree of luster shall be equal to the standards established in the DOD Standard Color Card of Official Standardized Shades for Heraldic Yarns for Embroidery and Ribbons, 1974.

3.3.3.3 Colorfastness. When tested as specified in 4.5.6, the dyed embroidery yarn shall show good fastness to wet-dry cleaning, laundering and light except the colorfastness for black, blue and red shades shall be not less than fair for laundering as defined in FED-STD-191.

3.3.4 Yarn and thread.

3.3.4.1 Bobbin yarn. The bobbin yarn used in the Schiffli embroidery shall be a continuous multi-filament nylon, a two-ply cotton yarn or a spun polyester yarn. The yarn shall be undyed except the yarn for Type I insignia may be dyed black. When the bobbin yarn is dyed black, the yarn shall show fair fastness to wet-dry cleaning and laundering as defined by FED-STD-191. The use of the sulphur dyes is prohibited. Testing shall be in accordance with 4.5.2.1.

3.3.4.2 Overedge stitching yarn and thread. The yarn and thread used for overedge stitching shall be 600 or 1200 denier multi-filament bright rayon for cover thread (see 3.5.7) and Type III, Ticket No. 00, 3 ply of A-A-52094 or Ticket No. 50, 2 ply of A-A-50199 for needle and looper threads.

3.3.4.3 Color. The color of the yarn and the thread used for overedge stitching shall be as specified for the embroidered edge on the applicable drawing or cartoon. The use of sulphur dyes is prohibited.

3.3.4.4. Colorfastness. When tested as specified in 4.5.6, the dyed overedge stitching yarn and thread shall show good fastness to wet-dry cleaning, laundering and light except the colorfastness for black, blue and red shades shall be not less than fair for laundering as defined in FED-STD-191.

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3.4. Design. The design and color for the background and device of the insignia shall be as shown on the original design drawing or applicable Government cartoon.

3.5 Construction. The insignia shall be embroidered on a Schiffli embroidery machine in accordance with this specification. When furnished by the Government, the applicable cartoon shall form a part of this specification. Designation of a specific embroidery span on a Government cartoon does not constitute a mandatory requirement, if all other requirements are met. Cartoons are furnished for the placement of different types of stitches. The cartoon will normally indicate the number of under, cover and miscellaneous stitches to produce an insignia as well as a mandatory requirement for total minimum stitches. The manufacturer may vary the under and cover stitches provided the total minimum stitches required is met and the insignia is equal to or better than the standard or guide sample. The density of cover stitches may be increased in critical areas where defects may occur. All thread ends shall be trimmed.

3.5.1 Embroidery stitches. The embroidery stitches shall be of the type, length, number and density specified on the applicable Government cartoon. The total stitches specified on the cartoon are full stitches and constitute two revolutions or movements of the machine. Underlay stitches shown on the cartoon by numbers are 1/2 stitches (one movement of the machine) but are converted to full stitches on the tabular listing. Except where otherwise indicated in the cartoon, all letters shall be underlaid and formed with blatt stitches. When required by contract provisions, the contractor shall submit a sample for approval. When a contractor's sample has been specifically approved for appearance of the embroidery stitching prior to production, subsequent deliveries shall by equal to or better than the approved sample.

3.5.2 Background stitches. Insignia requiring background stitching have either horizontal or vertical background stitching (as indicated on the applicable Government cartoon) and the stitch shall be 3/16 inch (0.48 cm) in length with a plus or minus tolerance of 1/32 inch (0.08 cm) interlocking in evenly spaced rows (satin fill). Horizontal background stitching shall run in a vertical row and vertical background stitching shall run in a horizontal row. The visible width of the embroidered row shall not be more than 5/64 inch (0.20 cm). The number of visual rows and the density (number of stitches per inch in a stitch line) shall be as indicated on the applicable Government cartoon. When a Government cartoon is not available, all Type I, Style 1 insignia shall have an embroidered background.

3.5.3 Device stitches. Unless otherwise specified, device stitches shall be in accordance with the applicable Government cartoon with respect to spacing the rows, number of visible rows and length and density of stitches. Translation of curvatures and other device features shall be executed with embroidery stitching finished evenly to effect a clean outline.

3.5.4 Underlay stitches. When underlay stitches are required, the number of stitches shall be as indicated on the cartoon. Where underlay is required in areas of design showing rows of stitches, the underlay stitches shall be inserted in a criss-cross pattern. In other areas requiring underlay, the underlay stitches shall be inserted in the direction most suitable for proper design execution.

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3.5.5 Stiele stitches. The length and density of required stiele stitches shall be tightly drawn and well executed to produce a clear, sharp edging.

3.5.6 Border stitches. All Type I insignia shall be bordered by a schrage stitch or a combination of a blatt stitch with a stiele stitch provided it is equal in density to the specified schrage stitch.

3.5.7 Overedge stitching. Unless indicated on the drawing that the insignia is not designed for overedge stitching, overedge stitching shall be superimposed over the border of all Type I, Style 1 insignia using yarn and thread specified in 3.3.4 as follows:

(a) The overedge stitching shall conform to stitch type 504 or 505 of FED-STD-751 except two ends of 600 denier or one end of 1200 denier shall be used for the cover threads. These shall be 24 stitches, plus or minus 4 stitches per inch and the stitch length shall be 1/8 inch (0.32 cm), plus or minus 1/32 inch (0.08 cm). Tolerance on width shall be acceptable only if the overedge stitching completely covers the border stitching when the specified border on the drawing is 1/8 inch (0.32 cm) in width.

(b) On circular or oval designs, there shall be a 1/2 inch (1.27 cm) overlap of the overedge stitching centered on the bottom of the insignia.

(c) The overedge shall start and finish at the bottom point for pointed shields.

(d) On rounds shield or similar designs with the bottom of the insignia round, the overedge shall start and finish in the upper left corner.

(e) Stitching for triangle, square, rectangular, polygonal, cylindrical or barrel shaped designs shall start and finish on any corner.

(f) Where practicable, overedge stitching shall be continuous from start to finish.

(g) Unless shown on the drawings, the rounding of corners or angles shall not be permitted.

(h) If stitching runs off the edge to begin a new direction for corners, the stitching ends shall be trimmed before being locked in by overedge stitching in the new direction.

(i) The final free end of the overedge stitching shall be brought to the back of the insignia and secured with adhesive or fused with a hot iron. The free end shall not be less than 1/2 inch (1.27 cm) in length and no adhesive shall be permitted within 1/8 inch (0.32 cm) of the finished insignia edge.

3.5.7.1 Optional methods of securing final free ends. At the option of the contractor, the final free end of the overedge stitching may be secured by blind-stitching 1/4 inch (0.64 cm) or less of the terminal end to the back of the insignia by hand or machine; or where size and configuration permit, the final end

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may be secured by application of a 3/8 inch (0.95 cm) by 1 inch (2.54 cm) (approximate) rectangle of masking tape. No part of the tape shall be within 1/8 inch (0.32 cm) of the finished edge of the insignia.

3.6 Stiffener. Type I insignia shall be treated on the reverse side with a fabric finish which will stiffen the insignia to facilitate subsequent processing. The fabric finish shall not affect the colors on the face of the insignia, cause puckering seriously affecting appearance of the insignia, and shall not react adversely to storage, laundering, wet-dry cleaning or ironing. The use of buckram specified in 3.3.2 shall be permitted provided it does not protrude at any point beyond the outline of the finished insignia. A clear plastic type backing fused into the material shall be permitted for use provided it is not heat transferable when pressed at temperatures up to 302 degrees Fahrenheit.

3.7 Cutting.

3.7.1 Type I (Organizational Insignia). Type I insignia shall be cut in single units and the base cloth shall be evenly trimmed as close to the outside of the embroidered edge as is possible without cutting the stitching. The base cloth shall be trimmed close enough to insure the overedge stitching covers the embroidered border when required.

3.7.2 Type II and III (Badges and Insignia, Branch of Service). Type II and Type III insignia shall be cut so a minimum of 5/8 inch (1.59 cm) base material is available between the design and raw edge.

3.8 Tolerances. Unless otherwise specified, the overall measurements of the finished insignia shall be within a tolerance of plus or minus 1/8 inch (0.32 cm) of the required measurements. Where no tolerances are cited, design dimensions shall have a tolerance of plus or minus 1/32 inch (0.08 cm).

3.9 Workmanship. The finished insignia shall be clean, well made and shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method documents or applicable paragraph(s) in the specification.

4.4 First article inspection. Inspection and testing of the first article shall be made on a completely fabricated insignia for all provisions of this specification applicable to the end product examination and tests (see 6.4).

4.5 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.5.1 Quality assurance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105 except when otherwise specified here after.

4.5.2 Component and material inspection. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials as specified herein.

4.5.2.1 Testing of components. Components and materials shall be tested as specified in Table I and FED-STD-191 unless otherwise indicated. The sample size for each characteristic of each component shall be as follows:

Lots Size (unit)	Sample Size
800 or less	2
801 to 22,000	3
22,001 to over	5

The lot shall be unacceptable if one or more units of any component fail to meet the requirements in Table I. The unit of expressing lot size and sample unit shall be as follows:

Component	Lot size expressed as	Sample size for testing
Buckram	Yard	1 linear yard full width
Embroidery yarn	Pound	1 holder for every color
Bobbin yard	Pound	5 bobbin for dyed yarn 1 bobbin for natural yarn
Cotton or polyester/cotton base cloth (as applicable)	Yard	3-1/2 yards full width

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When the data in the "# determ per S/U" and "Results" reported as" columns are not specified in Table I, they shall be as required in the referenced test method.

TABLE I Component Tests

Component	Characteristic	Regm't Para	Test Method	# determ Per S/U	Results Reported as
Embroidery yarn	Material identification	3.3.3 1/	ASTM-D-5034 1520	-	To nearest denier
	Ply	3.3.3	Visual	1	Pass or fail
	Denier	3.3.3	4021	1	To nearest denier
	Presence of Sulphur	3.3.3.1 1/	2020	1	Pass or fail
	Color 2/	3.3.3.1	Visual	1	Pass or fail
Base cloth	Material identification	3.3.1	1/	-	-
	Weave	3.3.1	Visual	1	Pass of fail
	Weight	3.3.1	5041	5	Nearest 0.1 oz/sq
	Ply	3.3.1	Visual	1	Pass or fail
	Breaking strength:				
	Warp(lbs)	3.3.1	ASTM-D-5034	5	Nearest 1 lb
	Filling(lbs)	3.3.1	ASTM-D-5034	5	Nearest 1 lb
	Shrinkage:				
	Warp	3.3.1	5550	3	Nearest 0.1%
	Filling	3.3.1	5550	3	Nearest 0.1%
	Colorfastness to laundering	3.3.1	5610	1	Pass or fail
Bobbin Yarn	Material identification	3.3.4.1 1/	1200 & 1530	1	Pass or fail
	Ply (when cotton yarn is used)	3.3.4.1	Visual	1	Pass or fail
	Color (when applicable) 2/	3.3.4.3	Visual	1	Pass or fail
	Presence of sulphur	3.3.4.3 1/	2020	1	Pass or fail
	Colorfastness (when applicable):				
	Wet-dry cleaning	3.3.4.4	5622	-	-
	Laundering	3.3.4.4	5614		

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TABLE I Component Tests - Continued

Component	Characteristic	Regm't Para	Test Method	# determ per s/u	Results reported as
Buckram	Weave	3.3.2	Visual	1	Pass or fail
	Weight	3.3.2	5041	5	Nearest 0.1 oz/ Square yard
	Yarn count:				
	Warp	3.3.2	5050	5	Nearest whole number
	Filling	3.3.2	5050	5	Nearest whole number
	Breaking strength:				
	Warp(lbs)	3.3.2	ASTM D 5034	5	Nearest 1 lb
	Filling (lbs)	3.3.2	ASTM D 5034	5	Nearest 1 lb
Overedge stitching yarn	Material identification	3.3.4.2	ASTM D 5034 or 1520	-	-
	Ply	3.3.4.2	Visual	1	Pass or fail
	Denier	3.3.4.2	4021	1	To nearest denier
	Presence of sulphur	3.3.4.3 1/	2020	1	Pass or fail
	Color	3.3.4	Visual	1	Pass or fail

1/ Unless otherwise specified, a contractor's certification of compliance may be accepted as evidence of conformance.

2/ When Government certified yarn is used, this test may be omitted by the contractor (see 6.5). The Government reserves the right to draw samples for verification.

4.5.3 Visual examination of the end item. Defects found during this examination shall be assigned demerit points from 1 to 3 as prescribed in Table III and added accordingly. All defects shall be as prescribed in Table III. Where two or more conditions are applicable for a single location, the highest point count applicable shall be scored. The lot size, sample size and sample unit shall be expressed in units of one finished insignia. The sample size and maximum sum of points permitted shall be as shown in Table II and expressed as maximum sum of points for 3 and 2 point defects combined and for all defects combined. The lot shall be unacceptable if the sum of the point value of the examined sample exceeds the limits established in Table II. Normal inspection distance for workmanship shall be at arm's length (16 to 22 inches) (40.64 cm to 55.88 cm). Examination for other defects may be determined at an inspection distance suitable for such examinations.

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TABLE II Maximum Sum of Points

Lot size	Sample	3 and 2 points combined	3, 2, and 1 points combined
Regular inspection			
Under 3200	125	25	35
3201 - 10,000	200	35	53
10,001 - 35,000	315	53	75
35,001 - 150,000	500	75	110
150,001 and up	800	110	150
Tightened inspection			
Under 3200	125	21	30
3201 - 10,000	200	30	45
10,001 - 35,000	315	45	68
35,001 - 150,000	500	68	103
150,001 and up	800	103	138

TABLE III. Defects

Examine	Defect	Point count		
Base cloth	Other than specified.....	3	-	-
	Any splice, cut, hole or tear visible on face	3	-	-
	Color shade not as specified.....	3	-	-
Stiffener/backing	Omitted or incomplete when required.....	3	-	-
	Back not coated with rubber base adhesive when required.....	3	-	-
	Coating not uniform or incomplete.....	-	-	1
	Buckram protrudes more than 1/32 inch (0.08 cm) beyond finished insignia.....	-	-	1
	Stiffener visible on front.....	3	-	-
	Stiffener/backing not as specified.....	3	-	-
	Design inadequately covering base cloth with background or underlay stitching showing through (thin or uneven stitch density).....	3	-	-

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TABLE II. Defects - Continued

Examine	Defect	Point count		
Construction (embroidery) General	Embroidery yarn not an acceptable match.....	3	-	-
	Background not as specified.....	3	-	-
	Device not as specified.....	3	-	-
	Background not interlocked in evenly spaced rows.....	-	-	1
	Spacing of rows not in accordance with cartoon.....	-	2	-
	Non-uniform embroidery stitch density.....	-	-	1
	Bobbin thread showing on face.....	-	2	-
	Tight tension-distorted surface of face side.	3	-	-
	Loose tension-loopy surface.....	-	*	-
	Traveling (walking) stitches on face of insignia.....	-	*	-
	Broken stitches or stitching incomplete (missing).....	-	*	-
Design	Any departure from cartoon or drawing.....	3	-	-
	Design or device poorly defined, sharp outline lacking.....	3	-	-
	Shift, device or outline misshaped, misplaced or overlapped incorrectly.....	-	*	-
	Device flat-underlay missing or not well rounded or prominent, underlay insufficient.	-	-	1
	Border not constructed or schrage stitches or a combination of blatt and stiele stitches when required.....	3	-	-
	Curvature or design features fuzzy, not clearly outlined.....	-	2	-
	Stiele stitches not firmly drawn to produce clearly defined rows or edges.....	-	2	-
	Stiele stitch in arcs, bars or device not as required by cartoon.....	-	2	-
Workmanship General	Kinked, bunched, tangled yarn on face.....	-	*	-
	Broken, scuffed, or cut yarns on face.....	-	*	-
	Spot, stain, stiffener, or foreign matter on face.....	-	-	1
	Two or more thread ends untrimmed.....	-	-	1
	More than two stumps longer than 1/16 inch (0.16 cm).....	-	-	1
	Creased or folded, will not flatten by manual pressure.....	-	*	-

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TABLE III. Defects - Continued

Examine	Defect	Point count		
	Embroidery yarn not an acceptable match when paired	3	-	-
Finish (border and edges)(Type I)	Omitted completely or not as specified	3	-	-
	Final free end not secured on back of insignia as specified	-	2	-
	Final free end not specified length	-	-	1
	Overedge does not adequately cover border	-	2	-
	Bunched or tangled yarns	-	2	-
	Broken or skipped stitches	-	2	-
	Raveled, ends not secured	-	2	-
	Run-off, not caught in base cloth	-	2	-
	Start and finish not at specified point	-	-	1
	Angles or pointed corners rounded off	-	-	1
	Tight or loose tension	-	-	1
	Overedge less tan 3/32 (0.24 cm) or more than 5/32 inch (0.40 cm) length	-	-	1
	Continuity lacking when design requires continuity	-	-	1
	Adhesive from final end exposed on finished edge	-	-	1
	Cut schrage or border stitches not adequately covered by overedge stitching	3	-	-
	Not finished as specified	3	-	-
Borders (base cloth) Types II and III	Not uniform in width or unevenly trimmed	-	2	-
	Corners or points not well defined	-	2	-
	Free material between edge design and raw edge of material not as specified	3	-	-
	Not cut in single units when required	3	-	-

* Defects designated by an (*) asterisk shall be classed as follows:
3 point defect when seriously affecting appearance or serviceability.
2 point defect when affecting appearance but not seriously.
1 point defect when affecting serviceability but not seriously.

4.5.4 Examination for dimensions. Measurements shall be taken of all points, lines, areas, curves and spaces for which dimensions are specified in the drawing, including:

Overall length and width
Border width
Base material around insignia

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Device location and dimension
Spaces and distances between intermediate points

The lot size shall be expressed in units of one insignia each for individual insignia. The sample unit shall be one completely fabricated insignia. Each measurement found to be outside of the specified tolerances shall be scored as a defect. The total defects found shall not exceed the maximum of the table in this paragraph. The sample shall be selected at random from among the units selected for visual examination. The tolerances for paragraph 3.8 or any other specific tolerances provided shall be applied.

Lot size	Sample size	No. of defects (Max)
Under 3200	5	0
3201 - 22,000	15	1
Over 22,000	25	22

4.5.5 Examination of preparation for deliver. An examination shall be made to determine the packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table IV. The sample unit shall be one shipping container fully prepared for delivery with the exception it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the Acceptable Quality Level (AQL) shall be 4.0 defects per hundred units.

TABLE IV. Classification of preparation for delivery defects

Examination	Defect
Markings, exterior and interior	Missing, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping,loose strapping or inadequate strapping. Bulging or distortion of containers.

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TABLE IV. Classification of preparation for delivery defects - Continued

Examination	Defect
Weight	Net weight exceeds requirements
Preservation (when applicable)	Omitted or not uniformly distributed in container.

4.5.5.1 Examination of count. An examination shall be made as prescribed in Table V to determine that the number of insignia per bundle, package, set-up box and shipping container is as required by section 5 of this specification.

TABLE V. Examination of count

Examine	Unit of product or lot expressed units of	Sample size	
		# Units in lot	# Units selected
Unit Packaging a. Number of insignia per bundle 1/	One bundle	up to 300 301 - 500 501 - 800 801 - 1300 1301 & over	5 10 15 25 30
Intermediate Packaging b. Number of bundles per set up box 1/	One set up box	up to 300 301 - 500 501 - 800 801 - 1300 1301 & over	5 10 15 25 30
Packing c. Number of set-up boxes per shipping container 1/	One shipping container	up to 3 4 to 6 7 & over	All 3 4

1/ The lot shall be unacceptable if any of the defects listed below are found:

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(a) Number of insignia per bundle. The average number of insignia per bundle is less than specified.

(b) Number of bundles per set-up box. The number of bundles in any set-up box is less than specified.

(c) Number of set-up boxes per shipping container. The total number of insignia in a shipping container is less than the total indicated on outside container marking.

(d) Number of insignia per shipping container. The number of insignia marked on the shipping container is more than the total of insignia marked on the packages in the container.

4.5.6 Testing of the end item. The end item shall be tested as specified in Table VI. The lot size shall be expressed in units of one insignia. The unit of product shall be eight (8) insignia. The lot shall be unacceptable if one or more units fail to meet the requirements referenced in Table VI. The sample size shall be as follows:

Lot size (each)	Sample size
22,000 or less	2
22,001 and over	3

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method
Embroidery and overedge Stitching yarn:		
Colorfastness to:		
Wet-dry cleaning	3.3.3.3 and 3.3.4.4	5622
Light	3.3.3.3 and 3.3.4.4	5660 1/
Laundering	3.3.3.3 and 3.3.4.4	5610 2/
Overedge stitching thread:		
Stitches per inch	3.5.7	5052 3/
Fabric finish:		
Plastic backing	3.6	4/

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1/ In the event of dispute resulting from the test with method 5660 or a result of anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with test method 5662.

2/ Only the stain on the fiber of the transfer cloth represented in the basic materials being tested shall be evaluated.

3/ The contractor shall furnish a certificate of compliance.

4/ Test will be accomplished by ironing the insignia against cloth material at required temperature. If plastic transfers to the cloth, the entire lot shall be unacceptable.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Insignia of one type, style and class shall be put up in bundles of twenty each and securely tied, or cross tied, with cotton twine or thread. Unless otherwise specified, insignia of one type, style and class may be placed in a clear polyethylene bag or blister pack. Closure of each bag shall be by heat-sealing or tape closure.

5.1.1.2 Intermediate packaging. Ten bundles (200 insignia) of one type, class and style preserved as specified in 5.1 shall be placed in a set-up paperboard box confirming to Type I, Variety 1, Class A, Style 3 or 4 of PPP-B-676. Closure shall be in accordance with the appendix of the container specification.

5.1.2 Commercial. Insignia shall be preserved and packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. Insignia preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to class weather-resistant, V3a or V3c of PPP-B-636. Closure shall be in accordance with the appendix of the applicable container specification. The weight of each shipping container shall not exceed the weight and size limitation of the container specification.

5.2.2 Level B. Insignia preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard box conforming to class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with the container specification. The weight of each shipping container shall not exceed the weight and size limitation of the container specification.

5.2.3 Commercial. Insignia of one type only, packaged as specified in 5.1 shall be packed in accordance with the applicable requirements of ASTM-D-3951.

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5.3 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type I, Class 1, Style 1, Size A, Wood Group I, II, III or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special markings required by the contract or purchase order, unit packages, intermediate boxes, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM-D-3951, as applicable.

5.4.1 Unit packaging. Unit packaging shall have the required information legibly printed or stamped in black on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. When unit packaging is secured by cotton twine or thread, the label shall be non-adhesive and secured under the tie. The bag or label shall indicate the following:

STOCK NUMBER
NOMENCLATURE
QUANTITY

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The insignia covered by this specification are intended to be worn by Department of Defense personnel to designate organization or skills of the wearer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type, style, class (see 1.2).

c. Drawing number (see 2.1.2).

d. When a guide sample will not be furnished (see 3.1).

e. When a first article is not required (see 3.2).

f. Cartoon (see 3.5).

g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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h. When palletization is required (see 5.3).

6.3 Samples. For access to guide samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed insignia. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. The invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Certification of embroidery yarn. Converters and suppliers of embroidery yarns may avail themselves of Government certification of their product with respect to shade matching, luster and colorfastness properties, once it is delivered into Government contractor's plant. Embroidery yarn which has not been so certified may cause delays in the manufacturing of insignia pending completion of Government certification tests. Inquiries about the certification procedures may be addressed to Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101-8419.

6.6 Subject term (key word) listing.

Branch of service insignia
Patches
Shoulder sleeve insignia

Custodians:
Army - IH
Air Force - 11

Review activities:
Air Force - 82,99
DLA - CT

User activity:
Air Force - 45

Preparing activity:
Army - IH

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER [illegible]-1-14632A	2 DOCUMENT DATE (YYMMDD) 920313

3 DOCUMENT TITLE

INSIGNIA, EMBROIDERED, ORGANIZATIONAL INSIGNIA AND BADGES, GENERAL SPECIFICATION FOR

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	f. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
The Institute of Heraldry	(703) 274-6636	284-6636
c ADDRESS (Include Zip Code) Bldg 15, Technical & Production Division Cameron Station Alexandria, VA 22304-5050	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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